

AI Persona Eval Report

VOICE QUALITY

Tested **6** Vapi Talk/PSTN-style calls covering intro, profile Q&A, GitHub Q&A, availability, booking, interruptions, and email confirmation. First-response latency was measured from Vapi's live latency panel and transcript timing; observed steady-state latency was **~840ms** after switching to GPT-4o mini, Deepgram Nova 3 English (India), and concise prompts. Transcription accuracy was manually labelled on **30** utterances with names, email spelling, slots, and interruptions; approximate word accuracy: **87%**. Booking task completion: **4/5** attempts successfully checked Google Calendar, proposed slots, created an event, and produced a Meet link.

CHAT GROUNDEDNESS

Built a golden set of **5** questions across resume, GitHub READMEs, recent commits, adversarial prompt injections, and unknown facts. Hallucination rate was measured by manual labels plus a judge pass requiring every factual claim to be supported by retrieved chunks: **0%**. Retrieval quality was measured by expected source files in top-5 chunks; precision@5: **0.80**, recall@5: **1.00**. Chat latency from the eval run was p50 **2766ms** and p95 **5158ms**.

FAILURE MODES

1. **Calendar invite failure**: service accounts cannot invite attendees without domain-wide delegation; fixed by switching to OAuth refresh-token booking.
2. **Vapi tool mismatch**: GET/POST schemas and response mappings differed from chat endpoints; fixed with dedicated voice endpoints and explicit tool schemas.
3. **Name/email transcription errors**: Indian names and email spelling were misheard; fixed with English (India), pronunciation guidance, and email confirmation before booking.

TRADEOFF

Chose **lower-latency voice responses** over **exhaustive spoken citations** because chat can keep richer citations, while voice needs short grounded answers and tool calls so barge-in and booking remain usable under the 2s first-response target.

TWO MORE WEEKS

Add repository-level code search, automated voice regression tests, richer calendar negotiation, production observability, and a larger labelled eval set.